

1	Find a vector of magnitude 6, which is perpendicular to each of the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$, where $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$.
2	$\vec{a}$ and $\vec{b}$ are two co-initial vectors forming the adjacent sides of a parallelogram such that $ \vec{a} = 10$, $ \vec{b} = 2$ and $\vec{a} \cdot \vec{b} = 12$. Find the area of the parallelogram.
3	A unit vector $\vec{a}$ is such that it makes an angle $\frac{\pi}{4}$ with x-axis, $\frac{\pi}{3}$ with y-axis and an acute angle θ with z-axis. Find θ and the components of $\vec{a}$.
4	Let two rods placed on the ground be represented by vectors $4\hat{i} - \hat{j} + 3\hat{k}$ and $-2\hat{i} + \hat{j} - 2\hat{k}$. Find a vector representing a flag-post of height 5 m that has to be erected perpendicular to both the rods.
5	Three honey bees were found flying along the vectors $\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}$, $\vec{b} = 4\hat{j} - 2\hat{k}$ and $\vec{c} = 3\hat{i} + 2\hat{k}$ respectively. Find the value of λ such that the path for $\vec{a} + \lambda \vec{b}$ is perpendicular to $\vec{c}$.
6	If A, B and C be three non-collinear points such that $\vec{AB} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{AC} = 2\hat{i} - 3\hat{j}$, then find the area of ΔABC .
7	Evaluate : $\int_{-4}^0 x+2 dx$
8	Find the intervals in which the function $f(x) = x^4 - 4x^3 + 4x^2 + 15$, is strictly increasing.

9	Find the interval(s) in which the function $f(x) = \frac{x}{\log x}$, where $x \in (0, 1) \cup (1, \infty)$, is increasing.
10	Determine the interval(s) in which $f(x) = 5x^{3/2} - 3x^{5/2}$, $x > 0$ is increasing.
11	Function f is defined as $f(x) = \begin{cases} 2x + 2, & \text{if } x < 2 \\ k, & \text{if } x = 2 \\ 3x, & \text{if } x > 2 \end{cases}$ Find the value of k for which the function f is continuous at $x = 2$.
12	Simplify : $\tan^{-1}\left(\frac{\cos x}{1 - \sin x}\right)$
13	Find value of k for continuity: $f(x) = \begin{cases} kx + 1, & x < 1 \\ 3, & x = 1 \\ 2x + k, & x > 1 \end{cases}$
14	Show that the relation S in set $\mathbb{R}$ of real numbers defined by $S = \{(a, b) : a \leq b^3, a \in \mathbb{R}, b \in \mathbb{R}\}$ is neither reflexive, nor symmetric, nor transitive.
15	If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$ find: $A + B$ AB BA - Verify whether $AB = BA$
16	Find the general solution of the differential equation $y \, dx - x \, dy + (x \log x) \, dx = 0.$

17	Find the particular solution of the differential equation $\frac{dy}{dx} - 3y \cot x = \sin 2x, \text{ given that } y = 2 \text{ when } x = \frac{\pi}{2}.$
18	Let $A = \mathbb{R} - \{3\}$ and $B = \mathbb{R} - \{1\}$. A function $f : A \rightarrow B$ is defined by $f(x) = \left(\frac{x-2}{x-3} \right).$ Find whether f is one-one and onto.
19	Four students of class XII are given a problem to solve independently. Their chances of solving the problem respectively are $\frac{1}{2}$, $\frac{1}{3}$, $\frac{2}{3}$ and $\frac{1}{5}$. Find the probability that at most one of them will solve the problem.
20	Find the point of intersection of the lines $\vec{r} = \hat{i} - \hat{j} + 6\hat{k} + \lambda(3\hat{i} - \hat{k}), \text{ and}$ $\vec{r} = -3\hat{j} + 3\hat{k} + \mu(\hat{i} + 2\hat{j} - \hat{k}).$ Also, find the vector equation of the line passing through the point of intersection of the given lines and perpendicular to both the lines.
21	Solve the following LPP graphically : Maximize $Z = 2x + 3y$ subject to the constraints $x + 4y \leq 8$ $2x + 3y \leq 12$ $3x + y \leq 9$ $x \geq 0, y \geq 0.$
22	Solve the following linear programming problem graphically : Minimise $Z = 6x + 7y$ subject to constraints $x + 2y \geq 240$ $3x + 4y \leq 620$ $2x + y \geq 180$ $x, y \geq 0.$

23	It is known that 20% of the students in a school have above 90% attendance and 80% of the students are irregular. Past year results show that 80% of students who have above 90% attendance and 20% of irregular students get 'A' grade in their annual examination. At the end of a year, a student is chosen at random from the school and is found to have an 'A' grade. What is the probability that the student is irregular ?
24	A builder wants to construct a dome shaped roof on a circular base as shown in the figure. The height of the dome is 6 m and the radius of the base is 10 m. The cross section of the dome is in the shape of a parabola.  Questions (i) Find the equation of the parabola representing the cross section of the dome if the vertex is at the top of the dome and axis is vertical. [2] (ii) Find the height of the dome at a point on the base which is 6 m away from the center. [3] (iii) Find the total surface area of the inner side of the dome. [5]

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A suspension bridge is supported by cables which are in the shape of a parabola. The cable hangs 25 m below the top of the pillars which are 100 m apart.

Questions

- (i) Find the equation of the cable if the lowest point is in the middle of the bridge. [2]
- (ii) Find the height of the cable at a point which is 30 m away from the center. [3]
- (iii) Find the length of the cable from one end to the other. [5]

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A biased die is tossed and respective probabilities for various faces to turn up are the following :

Face	1	2	3	4	5	6
Probability	0.1	0.24	0.19	0.18	0.15	K



Based on the above information, answer the following questions :

- (a) What is the value of K ? 2
- (b) If a face showing an even number has turned up, then what is the probability that it is the face with 2 or 4 ? 2

27	 A garden is made in the shape of a circle of radius 14 m. Two cross roads each of width 2 m pass through the center at right angles. Questions (i) Find the area of the paths. [2] (ii) Find the area of the lawn excluding the paths. [3] (iii) If the cost of grass is ₹30 per m^2 and the cost of paving the path is ₹50 per m^2, find the total cost of the garden. [5]
28	Using integration, find the area of the region bounded by the triangle ABC when its sides are given by the lines $4x - y + 5 = 0$, $x + y - 5 = 0$ and $x - 4y + 5 = 0$.
29	Find the value of a and b so that function f defined as : $f(x) = \begin{cases} \frac{x-2}{ x-2 } + a, & \text{if } x < 2 \\ a + b, & \text{if } x = 2 \\ \frac{x-2}{ x-2 } + b, & \text{if } x > 2 \end{cases}$ is a continuous function.
30	Find the absolute maximum and absolute minimum values of the function f given by $f(x) = \frac{x}{2} + \frac{2}{x}$, on the interval [1, 2].
31	Find : $\int \frac{2+\sin 2x}{1+\cos 2x} e^x dx$